

7.10 The incident wave can be expressed as

$$\psi_{inc}(x, t) = \frac{1}{\sqrt{2\pi}} \int A(\omega) \psi_{inc}(\omega, x, t) d\omega = \frac{1}{\sqrt{2\pi}} \int A(\omega) \exp\{ik(\omega)x - i\omega t\} d\omega$$

Then, the transmitted wave is

$$\psi_{trans}(x, t) = \frac{1}{\sqrt{2\pi}} \int A(\omega) \psi_{trans}(\omega, x, t) d\omega = \frac{1}{\sqrt{2\pi}} \int A(\omega) T(\omega) \exp\{ik(\omega)(x-L) - i\omega t\} d\omega$$

where $T(\omega) \equiv T(\omega) e^{i\phi(\omega)}$. Since $A(\omega)$ has narrow support near ω_0 , we can write

$$T(\omega) = T(\omega_0) \exp\left\{i\phi(\omega_0) + i \frac{d\phi}{d\omega}\bigg|_{\omega_0} (\omega - \omega_0)\right\},$$

and the transmitted wave becomes

$$\begin{aligned} \psi_{trans}(x, t) &= \frac{1}{\sqrt{2\pi}} \int A(\omega) T(\omega_0) \exp\left\{ik(\omega)(x-L) - i\omega t + i\phi(\omega_0) + i \frac{d\phi}{d\omega}\bigg|_{\omega_0} (\omega - \omega_0)\right\} d\omega \\ &= \frac{1}{\sqrt{2\pi}} T(\omega_0) \exp\left\{-i \frac{d\phi}{d\omega}\bigg|_{\omega_0} \omega_0\right\} \int A(\omega) \exp\left\{ik(\omega)(x-L) - i\omega\left(t - \frac{d\phi}{d\omega}\bigg|_{\omega_0}\right)\right\} d\omega \\ &= T(\omega_0) e^{i\beta} \psi_{inc}(x', t'), \end{aligned}$$

with $x' = x - L$, $t' = t - T = t - \left(d\phi/d\omega\right)|_{\omega_0}$, $\beta = -\omega_0 \left(d\phi/d\omega\right)|_{\omega_0}$.

7.10 (a) Using the result from Problem 7.2, we know that

$$E_t = \frac{2n e^{-i\omega d/c}}{2n \cos \tilde{z} - i(1+n^2) \sin \tilde{z}} E_i, \quad \tilde{z} = n\omega d/c$$

Then, the transmitted wave is given by

$$E_t \exp\{ikx - i\omega t\} = E_i \frac{2n}{2n \cos \tilde{z} - i(1+n^2) \sin \tilde{z}} \exp\{ik(x-d) - i\omega t\}.$$

Therefore, $T(\omega) = \frac{2n}{2n \cos \tilde{z} - i(1+n^2) \sin \tilde{z}}$. To achieve the (strange) result as presented in the problem, we can write $T(\omega)$ as

$$T(\omega) = \frac{2n e^{i\tilde{z}}}{(2n \cos \tilde{z} - i(1+n^2) \sin \tilde{z}) e^{i\tilde{z}}}.$$

$$\begin{aligned}
 \text{Since } e^{iz}(2n \cos z - i(1+n^2) \sin z) &= (\cos z + i \sin z)(2n \cos z - i(1+n^2) \sin z) \\
 &= 2n \cos^2 z + (1+n^2) \sin^2 z + i(2n \sin z \cos z - (1+n^2) \sin z \cos z) \\
 &= 2n \frac{1+\cos(2z)}{2} + (1+n^2) \frac{1-\cos(2z)}{2} - i \left((n^2+1) \frac{\sin(2z)}{2} - 2n \frac{\sin(2z)}{2} \right) \\
 &= \frac{1}{2}(n+1)^2 - \frac{1}{2}(n-1)^2 \cos(2z) - \frac{i}{2}(n-1)^2 \sin(2z),
 \end{aligned}$$

$$\text{then } T(w) = \frac{4n e^{iz}}{(n+1)^2 - (n-1)^2 \cos(2z) - i(n-1)^2 \sin(2z)}$$

Obviously, $T(w) = |T(w)| e^{i\phi(w)}$, with

$$|T(w)| = \frac{4n}{\left\{ [(n+1)^2 - (n-1)^2 \cos(2z)]^2 + [(n-1)^2 \sin(2z)]^2 \right\}^{1/2}}, \quad \phi(w) = z + \arctan \left[\frac{(n-1)^2 \sin(2z)}{(n+1)^2 - (n-1)^2 \cos(2z)} \right]$$

(b) Let us use the simpler expression, that $|T(w)| = \frac{2n}{[4n^2 \cos^2 z + (1+n^2)^2 \sin^2 z]^{1/2}}, \quad \phi(w) = \arctan \left[\frac{1+n^2}{2n} \tan z \right]$.

$$\text{Then, } \frac{d\phi}{dw} = \frac{1}{1 + \frac{(n^2+1)^2}{4n^2} \tan^2 z} \cdot \frac{1+n^2}{2n} \cdot \frac{1}{\cos^2 z} \cdot \frac{nd}{c} = \frac{2n(1+n^2)}{4n^2 \cos^2 z + (1+n^2)^2 \sin^2 z} \cdot \frac{nd}{c}.$$

$$\text{For } z \rightarrow 0 \text{ or } z = \pi, \quad |T(w)| = 1, \quad \frac{d\phi}{dw} = \frac{1+n^2}{2n} \cdot \frac{nd}{c}, \quad \text{or} \quad \frac{CT}{d} = \frac{1+n^2}{2}.$$

$$\text{For } z = \pi/2, \quad |T(w)| = \frac{2n}{1+n^2}, \quad \frac{d\phi}{dw} = \frac{2n}{1+n^2} \cdot \frac{nd}{c}, \quad \text{or} \quad \frac{CT}{d} = \frac{2n^2}{1+n^2}.$$

Not sure about the meaning of the average process.